Prompt Evaluation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Prompt Evaluation before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Structured Output and Schemas

Without opening the old lesson, explain one key idea from Structured **Output** and Schemas. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Examples and Rubrics

Without opening the old lesson, explain one key idea from Examples and **Rubric**s. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Tokens and Context

Without opening the old lesson, explain one key idea from Tokens and Context. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Prompt Evaluation trace and record the first step you can explain without help.

Use W2: Trace the worked case for Controlled prompt experiments. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Prompt Evaluation explanation against the supplied facts before discussing it.

Use the worked case for Controlled prompt experiments. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

Five validation tasks have pass results A:1,1,1,0,0 and B:0,1,0,1,1. Compare performance and explain what is required before choosing a prompt.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Prompt Evaluation target to a trace and a justified conclusion.

On four shared tasks, A **score**s 1,1,0,1 and B **score**s 1,0,1,1. Their totals tie. What should you inspect before choosing? Define a final evaluation step.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Prompt Evaluation case and show the deciding evidence.

On five fixed questions, A passes four and B passes three; B looks more attractive.

What conclusion is supported?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

Prompt A passes 8 of 10 tasks; B passes 9 of 10 but fails the only source-checking task. A source check is mandatory. Which can be accepted as-is?

My independent answer and reason

Write or draw your reasoning and evidence.

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Prompt Evaluation adds to Verified Study Buddy and keep the evidence.

Evaluate Study Buddy prompts on factual, ambiguous, missing-source, and adversarial questions.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

On four shared tasks, A **score**s 1,1,0,1 and B **score**s 1,0,1,1. Their totals tie. What should you inspect before choosing? Define a final evaluation step.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Prompt Evaluation facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Prompt Evaluation", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Trace the worked case

> Trace the worked case. Apply the supplied Prompt Evaluation facts and keep your reasoning.

On six development cases P passes cases 1, 2, 3, 4; Q passes 1, 2, 3, 5, 6. Missing-approval cases are 4 and 6. Compare totals and condition results. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Audit the limit

> Audit the limit. Apply the supplied Prompt Evaluation facts and keep your reasoning.

A report omits this qualification: Prompt selection on development cases cannot be reported as independent final performance. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Prompt Evaluation facts and keep your reasoning.

On eight **paired case**s A passes 1–5; B passes 1–4, 6, 7. Special cases are 5 and 8. Compare overall and special-case results; select by total passes.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Prompt Evaluation facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Prompt Evaluation", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Prompt Evaluation

> A fictional Study Buddy compares two prompts across six development questions. Explain the labelled step before checking the changed condition.

A fictional Study Buddy compares two prompts across six development questions.

A passes cases 1,2,3,4. B passes 1,2,5,6. Cases 5 and 6 require refusing unsupported claims.

Mandatory selection gate: pass both refusal cases. All cases use fixed sources and one common **rubric**.

AI ACADEMY / AI-3/9

Same total, different mandatory-case results

	Total passes /6	Refusal passes /2
Prompt A	4	0
Prompt B	4	2

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-3/9

Prompt Evaluation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Paired total

A and B both pass 4/6.



2. Different failures

A alone passes 3,4; B alone passes 5,6.



3. Mandatory slice

A passes 0/2 refusal cases; B passes 2/2.



4. Selection

B clears the declared gate, but its failures 3,4 remain visible.

CHANGE ONE THING

Remove the gate and retain total passes as the only criterion.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why is B′s selection defensible here without calling it universally superior?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Use Study Buddy case matrices with explicit priorities, source support gates and fresh final questions.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use Study Buddy case matrices with explicit priorities, source support gates and fresh final questions.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional Study Buddy compares two prompts across six development questions.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? B clears the declared gate, but its failures 3,4 remain visible.

1. Different failures

2. Paired total

3. Mandatory slice

4. Selection

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Selection B clears the declared gate, but its failures 3,4 remain visible. Equal totals do not imply equal behavior.

The predeclared gate expresses an important requirement instead of choosing a criterion after seeing the result. Selecting B is justified under that rule, not by claiming it is best on every question.

After development, freeze the chosen prompt and test new questions, including both ordinary and unsupported cases.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Remove the gate and retain total passes as the only criterion. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. The record is a tie; another predeclared tie policy or further development **evidence** is needed.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The record is a tie; another predeclared tie policy or further development **evidence** is needed. Both prompts have four passes on six questions.

Repair a missing building block

Review Structured **Output** and Schemas (ai-3/8). Goal: Separate JSON parsing, schema validation and source support checks. Short practice: Which object demonstrates that valid JSON and valid types are insufficient for truth?.

Return to this week and explain what you changed.

Review Examples and **Rubric**s (ai-3/6). Goal: Write a prompt specification with observable response criteria. Short practice: Explain why neither original nor changed draft is a full pass, and provide a complete repaired card..

Return to this week and explain what you changed.

Review Tokens and Context (ai-3/3). Goal: Account for fixed text, **evidence** and reserved **output** under a toy budget. Short practice: Can A alone justify answering 20 seats after the exception is known?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

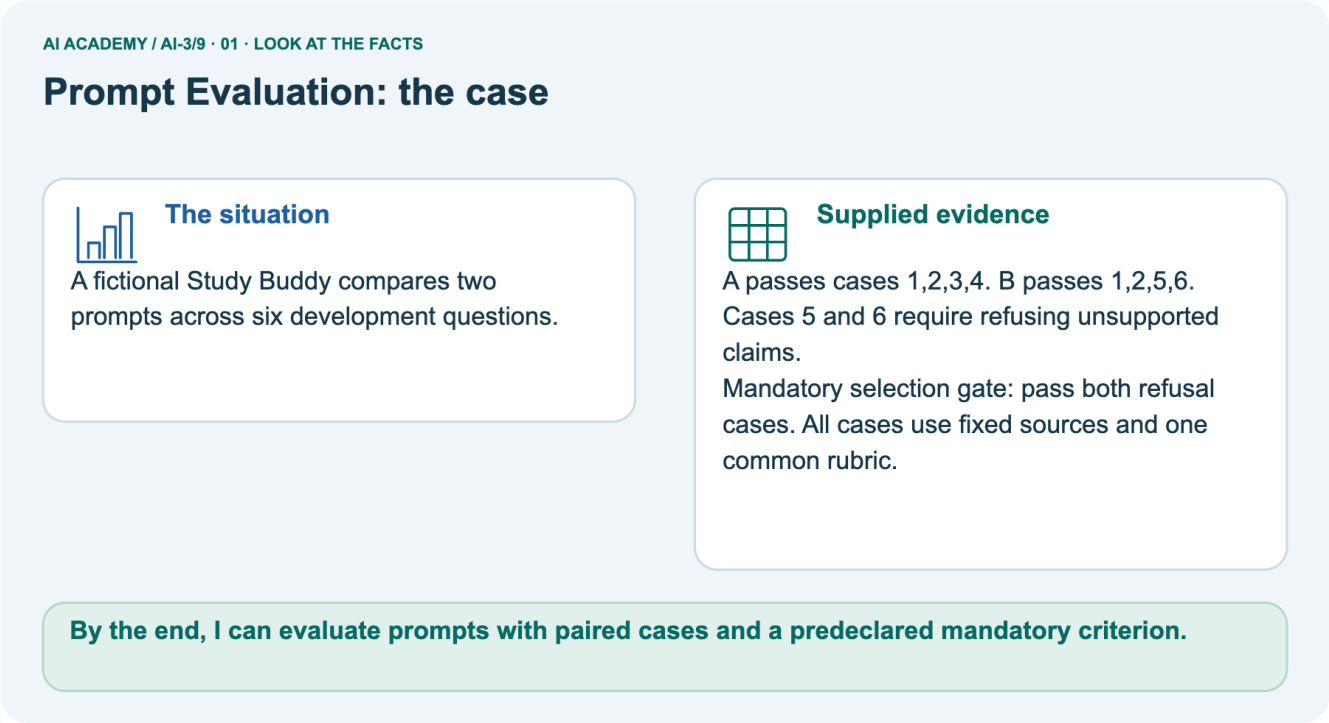
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Prompt Evaluation picture from its labels, then record a relevant limit.

01. Prompt Evaluation: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

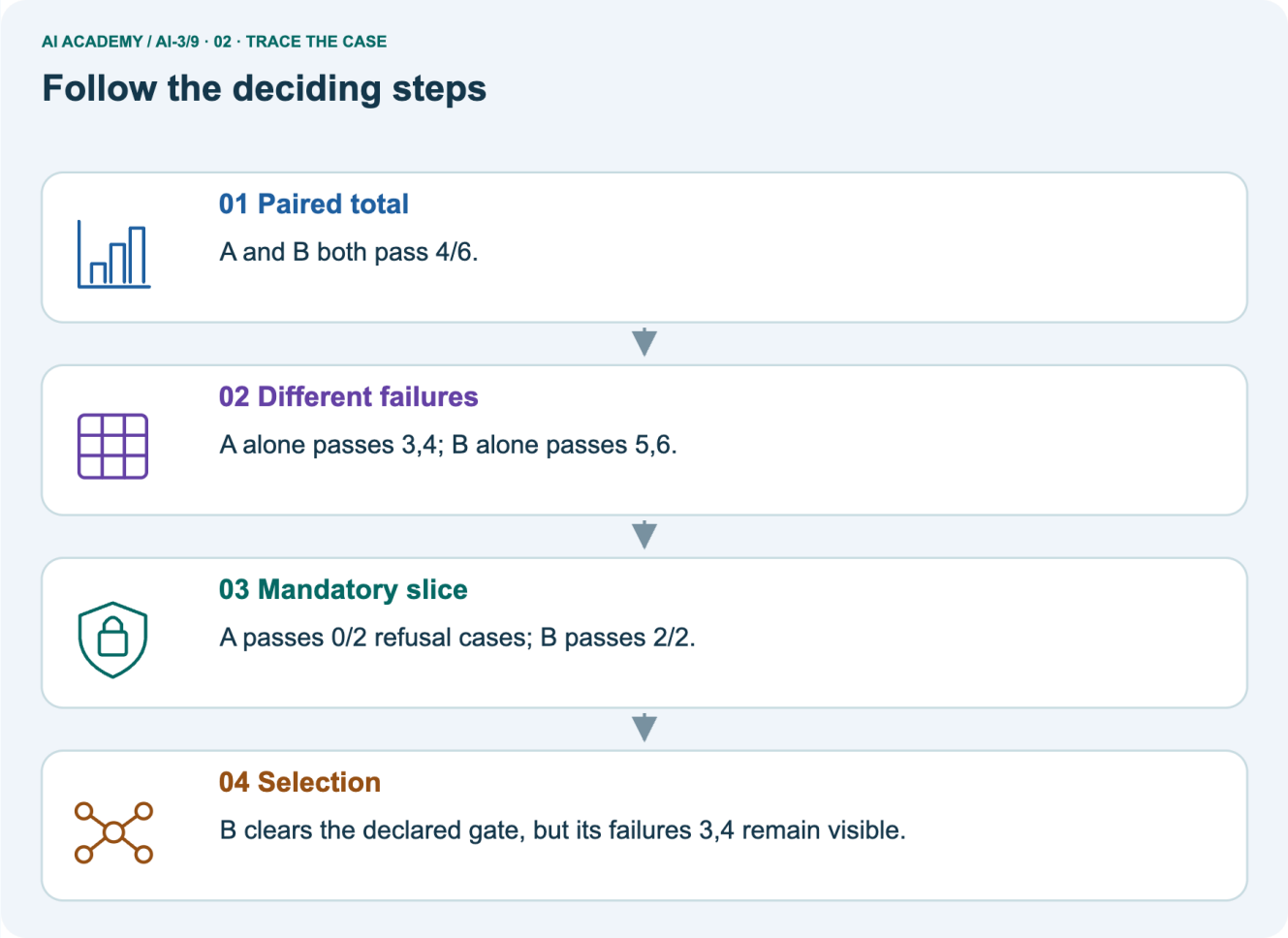
A fictional Study Buddy compares two prompts across six development questions. Supplied facts: A passes cases 1,2,3,4. B passes 1,2,5,6. Cases 5 and 6 require refusing unsupported claims.

Mandatory selection gate: pass both refusal cases. All cases use fixed sources and one common **rubric**.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



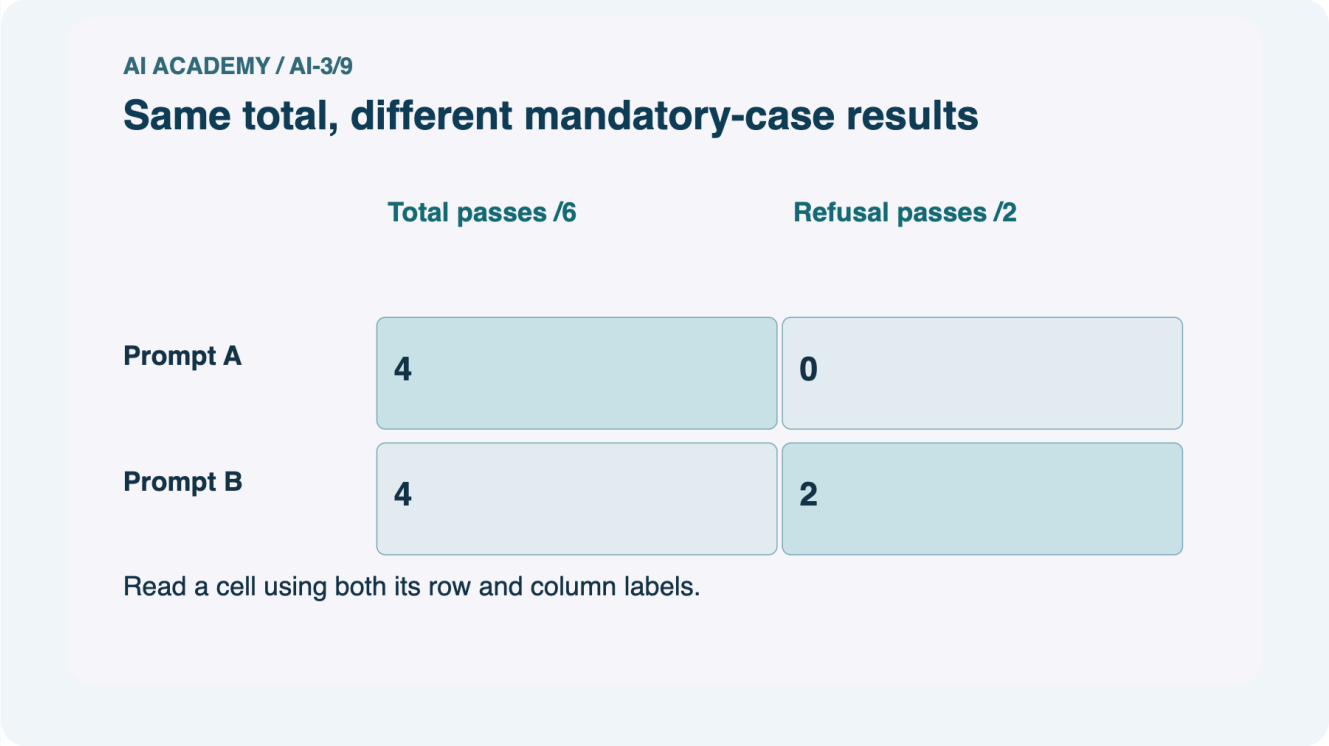
Follow the labelled picture with the matching explanation. Open the vector diagram

Paired total: A and B both pass 4/6. → Different failures: A alone passes 3,4; B alone passes 5,6. → Mandatory slice: A passes 0/2 refusal cases; B passes 2/2. → Selection: B clears the declared gate, but its failures 3,4 remain visible.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Same total, different mandatory-case results



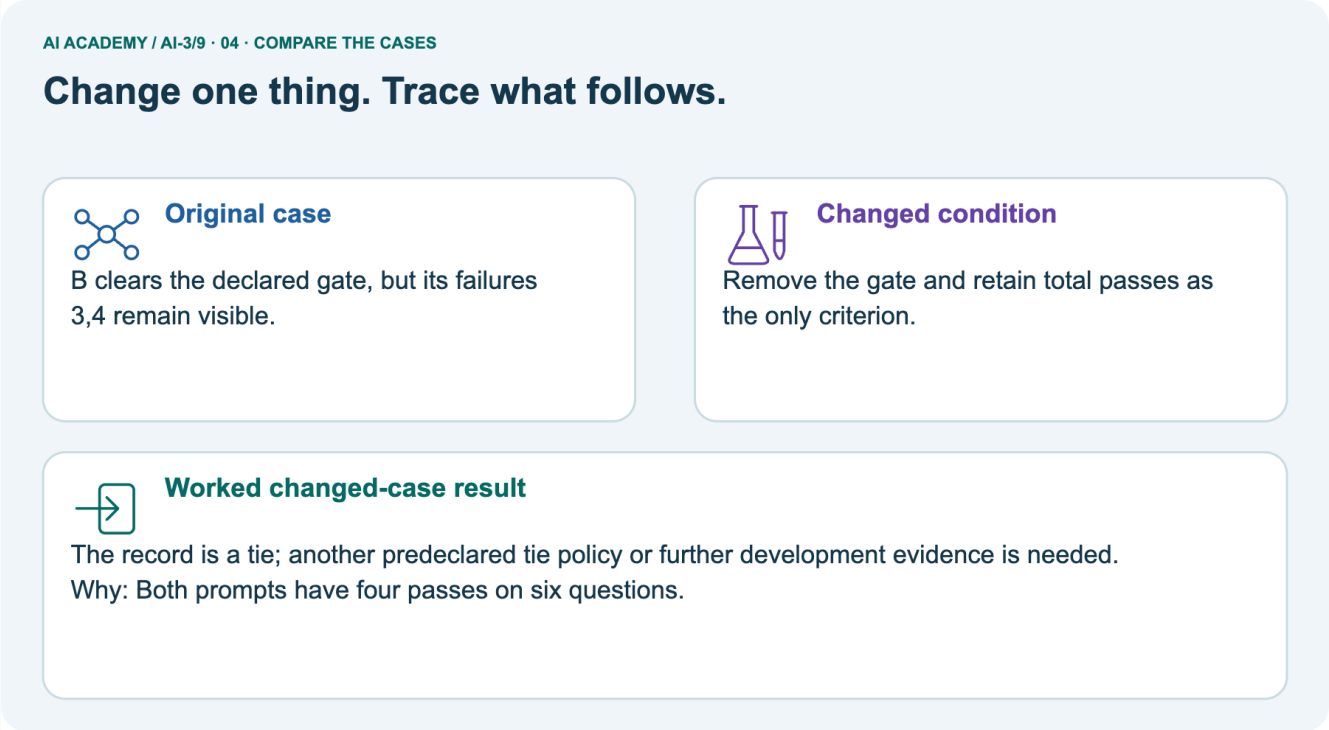
Follow the labelled picture with the matching explanation. Open the vector diagram

Same total, different mandatory-case results. Prompt A: Total passes /6 = 4, Refusal passes /2 = 0. Prompt B: Total passes /6 = 4, Refusal passes /2 = 2.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

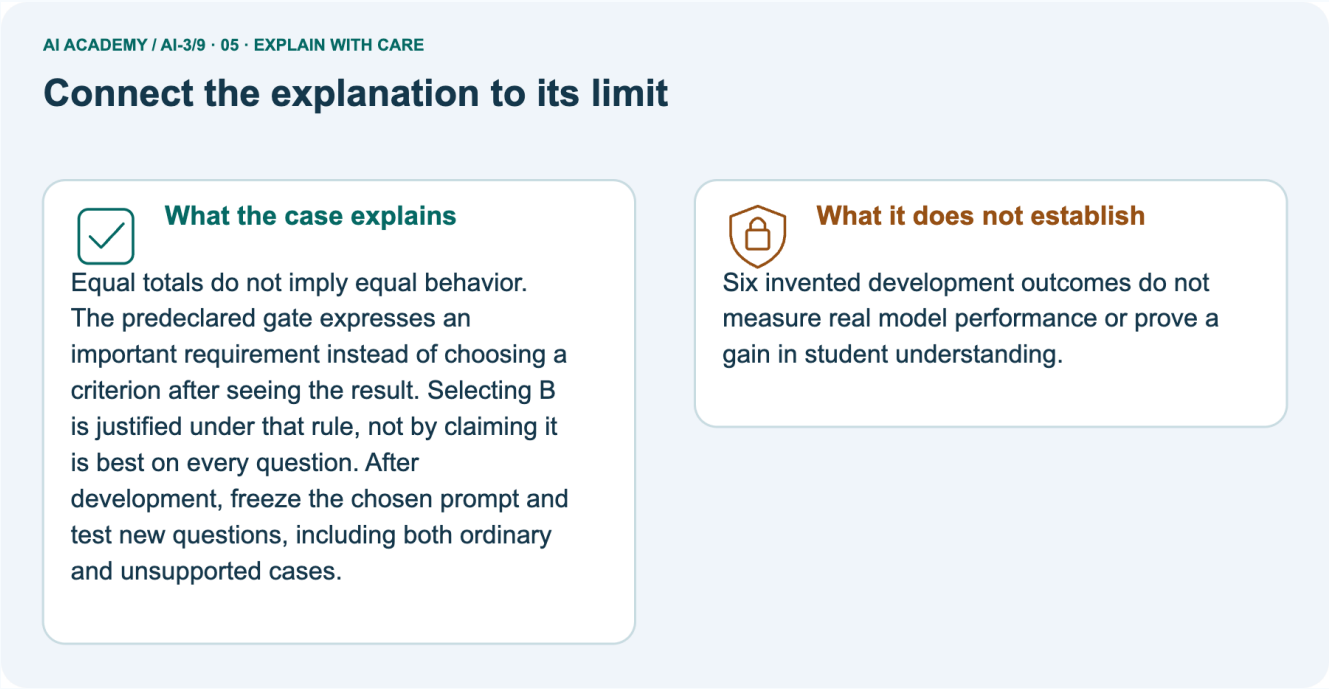
Original case: B clears the declared gate, but its failures 3,4 remain visible. Changed condition: Remove the gate and retain total passes as the only criterion.

Expected result: The record is a tie; another predeclared tie policy or further development **evidence** is needed. Reason: Both prompts have four passes on six questions.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Equal totals do not imply equal behavior. The predeclared gate expresses an important requirement instead of choosing a criterion after seeing the result.

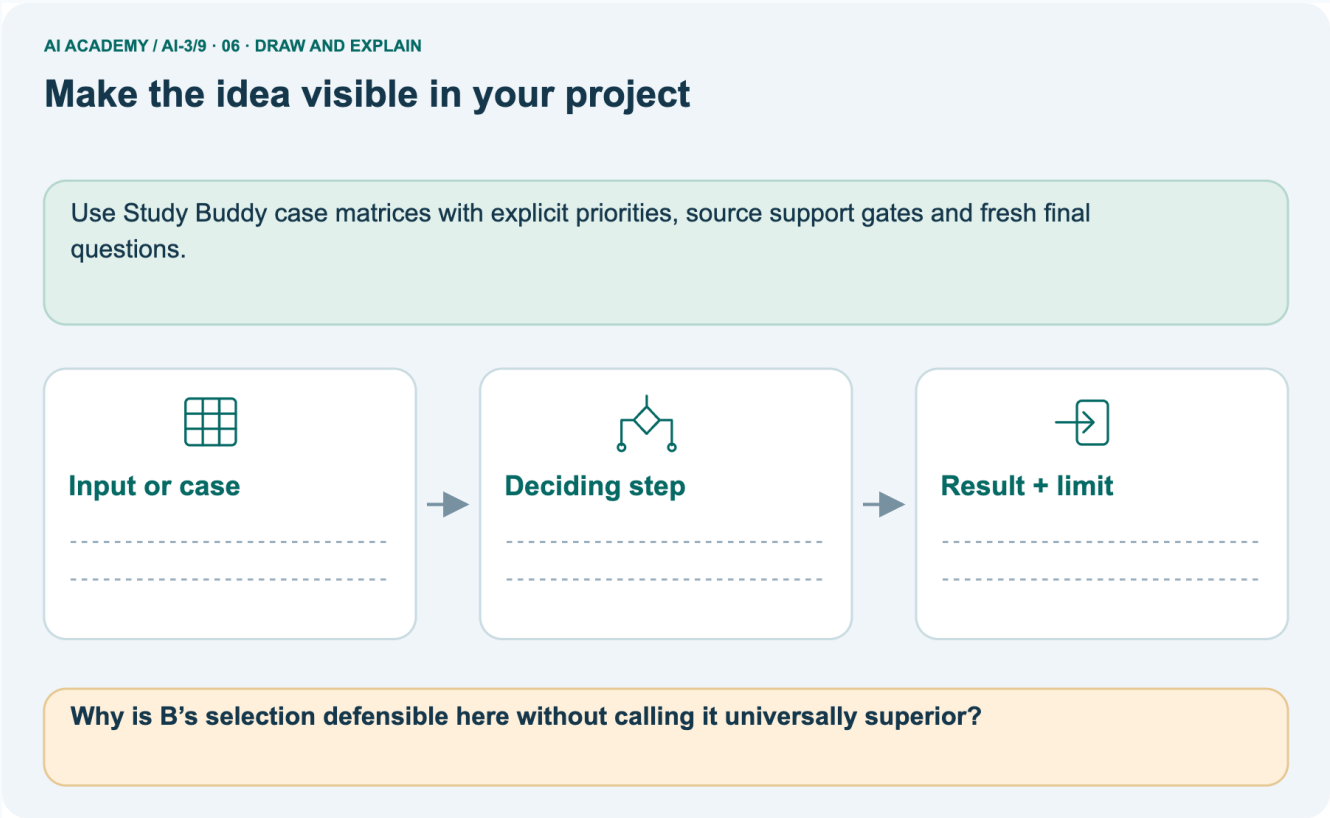
Selecting B is justified under that rule, not by claiming it is best on every question. After development, freeze the chosen prompt and test new questions, including both ordinary and unsupported cases.

Limit: Six invented development outcomes do not measure real model performance or prove a gain in student understanding.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use Study Buddy case matrices with explicit priorities, source support gates and fresh final questions. Draw your **input**, deciding step, result and limit.

Supported question: Why is B’s selection defensible here without calling it universally superior?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.